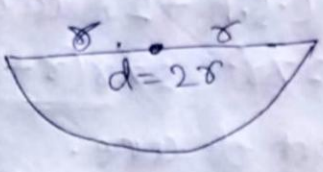
(14) ③ Circle & Semi Circle:-



'r' → radius
d - diameter = 2r

Circumference / Perimeter = $2\pi r$

Area = πr^2

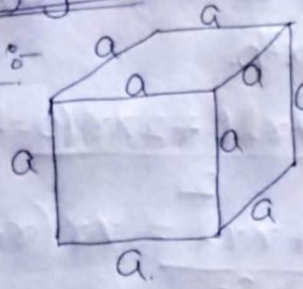


Circumference = $\pi r + 2r$
$= r(\pi + 2)$

Area = $\frac{\pi r^2}{2}$

(11) 3D figures:- Mainly 5 figures:-

1) Cube:-



↳ All sides & angles are equal.
↳ longest diagonal means which passes through the center & its '3d' diagonal.

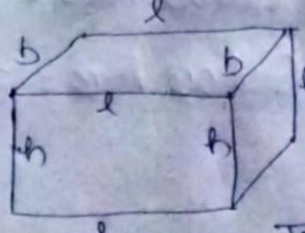
Volume = Base area x height = $a^2 \times a = a^3$

longest diagonal = $\sqrt{a^2+a^2+a^2} = \sqrt{3}\,a$

Lateral Surface Area = Excluding top & bottom remaining area
= 4 Area of square [4 faces] = $4a^2$

Total Surface Area = Area of all faces
= 6 faces area = $6a^2$

2) Cuboid:- Opposite sides & angles are equal.



Volume = Base Area x height = $lb \times h = lbh$

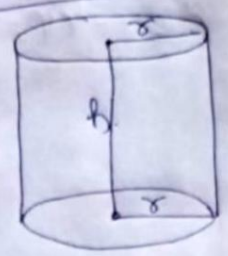
longest diagonal = $\sqrt{l^2+b^2+h^2}$

Lateral Surface Area = $2h(l+b)$

Total Surface Area = $2(lb+bh+hl)$

3) Cylinder:- Volume:- Base Area x height

$= \pi r^2 \times h = \pi r^2 h$



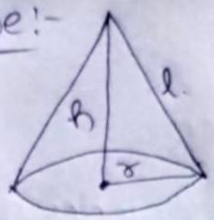
Lateral Surface Area:- Circumference of 'b' height.

$= 2\pi r \times h = 2\pi rh$.

Total Surface Area = L.S.A + 2 Area of circle

$= 2\pi rh + 2\pi r^2 = 2\pi r(r+h)$

4) Cone:-



'l' is slant height of cone.

$V_{cyl} = 3\ V_{con}$ when h & r equal.

So, Volume of Cone = $V_{con} = \frac{1}{3} V_{cyl}$

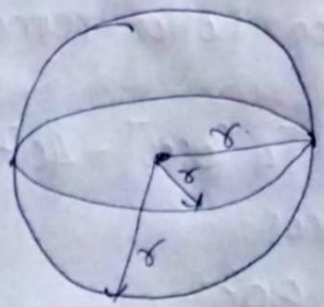
$V = \frac{1}{3}\pi r^2 h$.

Lateral Surface Area = $\frac{2\pi rl + 2\pi r(0-l)}{2} = \pi rl$.

Total Surface Area = $\pi r^2 + \pi rl = \pi r(r+l)$

Slant height = $l = \sqrt{r^2+h^2}$

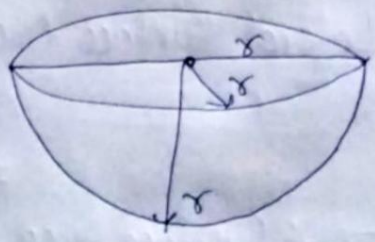
5) Sphere & Hemisphere:-



$V = \frac{4}{3}\pi r^3$

Total Surface Area

$= 4\pi r^2$



$V = \frac{2}{3}\pi r^3$

Lateral Surface Area = $2\pi r^2$

Total Surface Area = $3\pi r^2$

Ex:-1 [B.Q:-15]:- Find the cost of flooring the class room @ 12/- ~~m²~~. If the length & breadth of class room are in the ratio 3:2 & its perimeter is 40 m?

Ans:- 'P' of □ = $2(l+b) = 2(3x+2x) = 40$

$\Rightarrow 10x = 40 \Rightarrow x = 4$ m.

$l = 3x = 12$ m, $b = 2x = 8$ m



flooring means face means Area $= lb = 12 \times 8 = 96\ m^2$

Cost of flooring $= 96\ m^2 \times 12 = 96 \times 10 + 96 \times 2$

$= 1152/-$

Ex:-2:- [B.Q:-16]:- How many squares with sides of 2cm cover the surface of a rectangle with a length of 24cm & breadth of 8cm?

Ans:- No. of squares in □ plot $= \frac{\text{Area of } \square}{\text{Area of each } \square} = \frac{lb}{a^2}$

$= \frac{\overset{12}{\cancel{24}} \times 8}{\cancel{2} \times \cancel{2}} = 48$ squares we can plot.

Ex:-3:- [B.Q:-17]:- The circumference of a front wheel of a cart is 30 feet long & that of back wheel is 36 ft long what is the distance travelled by the cart, when the front wheel has done five more revolutions than the rear wheel?

Ans:- The cart covering the dist means how much front wheel covered the dist which is same as back wheel cover.

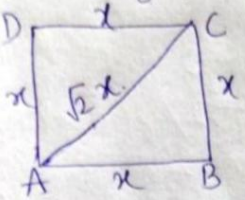
~~dist = Speed × time~~. Let's consider front wheel revolutions 'n'.

dist covered $\Rightarrow n \times 30 = (n-5)36 \Rightarrow 30n = 36n - 180$

$\Rightarrow 6n = 180 \Rightarrow n = 30$ revolutions

dist covered $= n \times 30 = 30 \times 30 = 900$ feet.

Ex-3:- [B.Q:-19]:- A person walked diagonally across a square plot. Approx, what was the % saved by not walking along the edge?



If walking through sides it's taking $x+x=2x$ distance [AC [AB+BC]].

But walking diagonally takes $\sqrt{2}x = 1.414x$

% change (saved dist %) $= \frac{2x-1.414x}{2x} \times 100$

$= \frac{0.5886x}{2x} \times 100 \simeq$ **30%** Approx

Ex-4:- [B.Q:-20]:- The length of rectangular plot is 20 meters more than its breadth. If the cost of fencing the plot is 26.5/- per meter is 5300/-, what is the length?

Ans:- Lets Assume l as l & $b = l-20$

fencing of Rectangle = Perimeter $= 2(l+l-20) = 4l-40$

fencing amount $= (4l-40)26.5 = 5300$

$\Rightarrow 106l - 1060 = 5300 \Rightarrow 106l = 6360$

$\Rightarrow l = \frac{6360}{106} =$ **60 meters**

Ex-5:- A cycle wheel rotates 5000 revolutions to cover 10km. Find radius, circumference & Area?

Ans:- 5000 revolutions = 5000 × Circumference

$\Rightarrow 5000 \times 2\pi r = 10000$ meters $\Rightarrow$ **$r = \frac{1}{\pi}$ m**

Circumference $= 2\pi r = 2 \times \pi \times \frac{1}{\pi} m =$ **2 m**

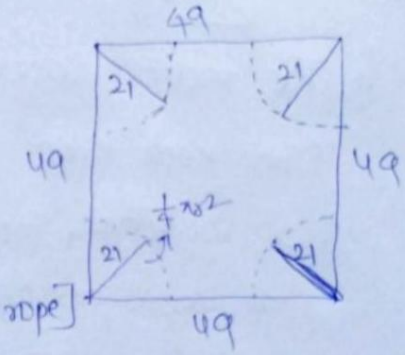
Area $= \pi r^2 = \pi \times \frac{1}{\pi} \times \frac{1}{\pi} m^2 = \frac{1}{\pi} m^2$

Area $= \frac{7}{22} m^2$

Ex:6:- In a Square shaped grass field 4 cows are tethered at 4 corners with ropes of length 21 meters. If side of land is 49 m then

(i) Area of grass grazed by cows?

(ii) Area of Ungrazed portion?



Ans:- Area of land = $49^2 = 2401\ m^2$

(i) Area of grazed portion =

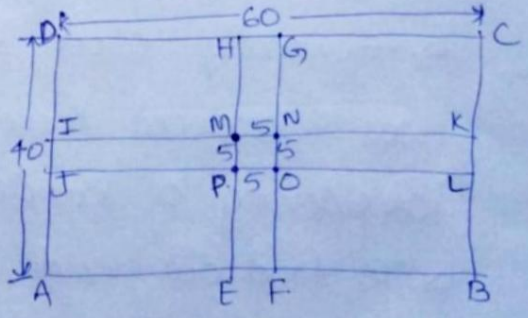
$= \text{Total Area} = 4 \times \frac{1}{4}\pi r^2$ [r = length of rope]

$= \frac{22}{7} \times \frac{3}{21} \times 21 = 1386\ m^2$

(ii) Area of Ungrazed portion:- Total Area - Grazed Area

$= (2401 - 1386)\ m^2 = 1015\ m^2$

Ex:7:- In a rectangular park of dimensions 60×40m in the middle 2 roads of width 5m in such a way that one is ||el to length & the other ||el to breadth. Then the Area of roads?



Ans:- Roads Area = □EFGH + □IJKL − □MNOP

$= 60 \times 5 + 40 \times 5 - 5 \times 5$

$= 300 + 200 - 25$

$= 475\ m^2$

Ex:8:- 3 small cubes of sides 6m, 8m & 10m are melted to form a bigger cube. Side of Bigger cube is?

Ans:- Volume $_{Big} = V_{s_1} + V_{s_2} + V_{s_3} \Rightarrow S^3 = s_1^3 + s_2^3 + s_3^3$

$S^3 = 6^3 + 8^3 + 10^3 = 216 + 512 + 1000 = 1728\ m^3$

$S = \sqrt[3]{1728\ m^3} = \boxed{12\ m}$